

Document Information

Project Name	Mota50: Integrated Fleet Management & Smart Driver Digital Log Application		
Project Support	World Vision: Global Technology and Digital Solutions		
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List of Abbreviations

AI	Artificial Intelligence
API	Application Programming Interface
ERP	Enterprise Resource Planning
FMS	Fleet Management System
GPS	Global Positioning System
IoT	Internet of Things
IoV	Internet of Vehicles
MFA	Multi-Factor Authentication
ML	Machine Learning
NGO	Non-Governmental Organisation
OBD	On-Board Diagnostics
RBAC	Role-Based Access Control
SDG	Sustainable Development Goal
SUC	Secret Unknown Cypher
TOGAF	The Open Group Architecture Framework
UI	User Interface
WVZ	World Vision Zambia

Instructions

DemoVideo	< https://youtu.be/WD6Edo9xVHQ >	
Prototype	< https://wvz-mota50.onrender.com/ >	
Role	User	Password
admin	admin@wvz.org	password123
driver	driver@wvz.org	password123
non-driver	staff@wvz.org	password123
finance	finance@wvz.org	password123
MFA Code	123456	

The table shows the mock user accounts used to access different roles in the Moto50 prototype, each with simple test credentials for demonstration. The MFA code is a universal verification code used to simulate two-factor authentication during login.

Introduction

Moto50 is an innovative fleet management and smart driver-log system designed to support World Vision Zambia's operational needs. The name combines "mota," a Nyanja term for car, with "50," representing the year 1950 when World Vision was founded, symbolising both movement and legacy. The project aims to digitise vehicle bookings, trip logging, inspections, and cost-centre reporting, offering a modern, reliable, and transparent platform tailored to the organisation's humanitarian mission.

Literature Review

The reviewed studies collectively highlight the global shift toward integrated, data-driven fleet management systems, yet they also reveal critical gaps when applied to humanitarian operations such as those of World Vision Zambia. While existing research demonstrates the value of telematics, unified platforms, and strong security frameworks, most solutions are designed for commercial fleets in high-connectivity environments. They generally overlook NGO-specific needs such as cost-centre accountability, donor-compliant reporting, rural deployment constraints, and role-based workflows. These gaps create a clear opportunity to design a context-aware system tailored to WVZ's operational realities. Our proposed solution directly targets these unmet needs by prioritising simplicity, transparency, and field-readiness.

TABLE I. Summary of Literature and Gaps in the Study.

No	Authors	Title	Key Findings	Methods	Proposed Solutions	Gaps
[1]	Wedha et al., 2023	Enterprise Architecture for Fleet Management	Integrated architecture improves efficiency via IoT, data analytics.	Enterprise-architecture modelling (TOGAF/ArchiMate), system analysis.	Enterprise fleet system with tracking, maintenance & driver modules.	Not NGO-focused; no cost-centre tracking, donor reporting, or offline capability.
[2]	Nyati, 2023	Integrated Monitoring & Maintenance	Fragmented tools cause inefficiency; integration improves workflows.	Workflow analysis, system design, UI mock-ups.	Unified diagnostics, tracking & maintenance portal.	Lacks role-based workflows, project linking, and rural/offline support. Commercial focus only.
[3]	Victor, 2025	Fleet Telematics Review	Telematics improves safety, fuel use & predictive maintenance.	Narrative review of telematics tech & performance metrics.	Data-driven telematics framework (GPS, OBD, behaviour analytics).	Requires advanced connectivity/hardware; not adapted to low-resource NGO environments.
[4]	Faridi et al., 2021	Cloud IoV Fleet Weight System	IoV + cloud enhances tracking, load monitoring & compliance.	System description, cloud/IoV architecture review.	Cloud-based IoV fleet weight & route management.	Focus on cargo/weight; not aligned to NGO bookings, inspections, or cost allocation.
[5]	Hamadaqa et al., 2020	Security for Fleet Management	FMS is vulnerable to impersonation & falsified data.	Security analysis, proposal of SUC-based architecture.	Clone-resistant identities + secure authentication.	Hardware-heavy & impractical; need lightweight security (MFA, hashed logs).

Methodology

Project Objectives

- I. To develop an integrated digital fleet management system that streamlines vehicle bookings, trip logging, and inspection workflows for drivers and staff across World Vision Zambia.
- II. To enhance accountability and donor-compliant reporting by implementing cost-centre allocation, audit-ready logs, and real-time dashboards for fleet utilisation, operational costs, and compliance metrics.
- III. To design a lightweight, field-ready architecture that supports low-connectivity environments, secure authentication, and optional telematics modules tailored to WVZ's technical and operational constraints.

Software Process Model

We use the Spiral Model because it allows iterative development and ongoing risk management for WVZ's complex and evolving requirements. Each cycle includes refining needs, building a prototype, and gathering user feedback.

Prototypes will progress from bookings and trip logs to inspections, compliance, and dashboards, with optional telematics added later. This ensures the system stays practical, user-driven, and aligned with WVZ's mission.

Architectural Diagram

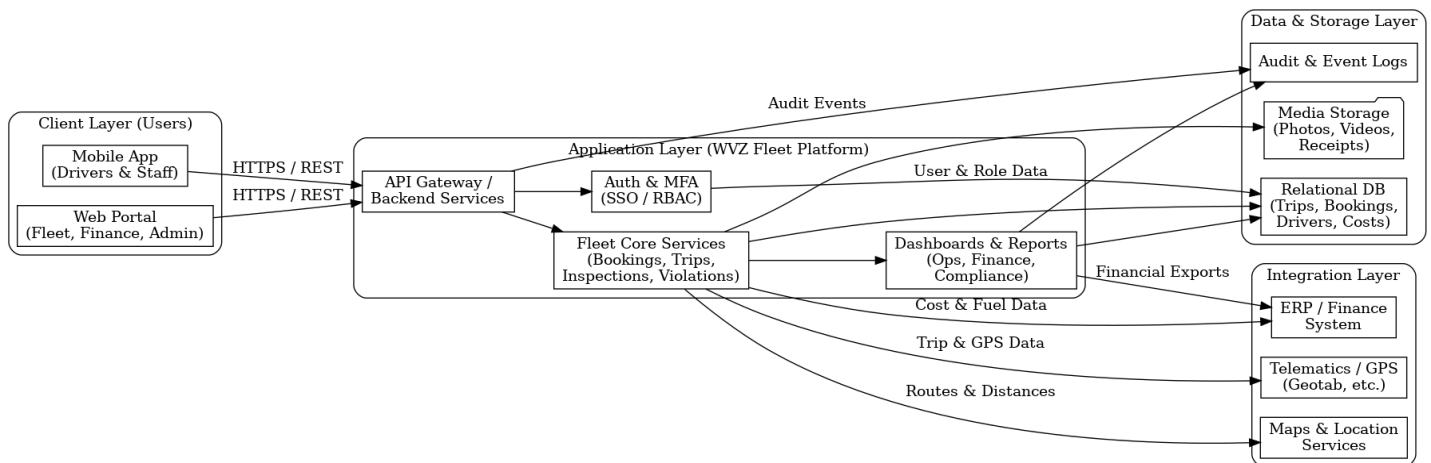


Figure 1. The diagram illustrates the system's core layers, client apps, backend services, integrations, and data storage. It highlights a modular structure designed to support unified fleet operations in low-resource environments.

Stack

For the prototype, we used **Vanilla HTML**, **Tailwind CSS** and **JavaScript** hosted on **Render** to demonstrate the system functionality. We propose a **React/React Native frontend**, **Node.js backend**, and **PostgreSQL database** with cloud storage for media. This stack is lightweight, scalable, offline-friendly, and integrates easily with telematics and ERP systems—ideal for both prototype and full deployment.

Use Case Diagram

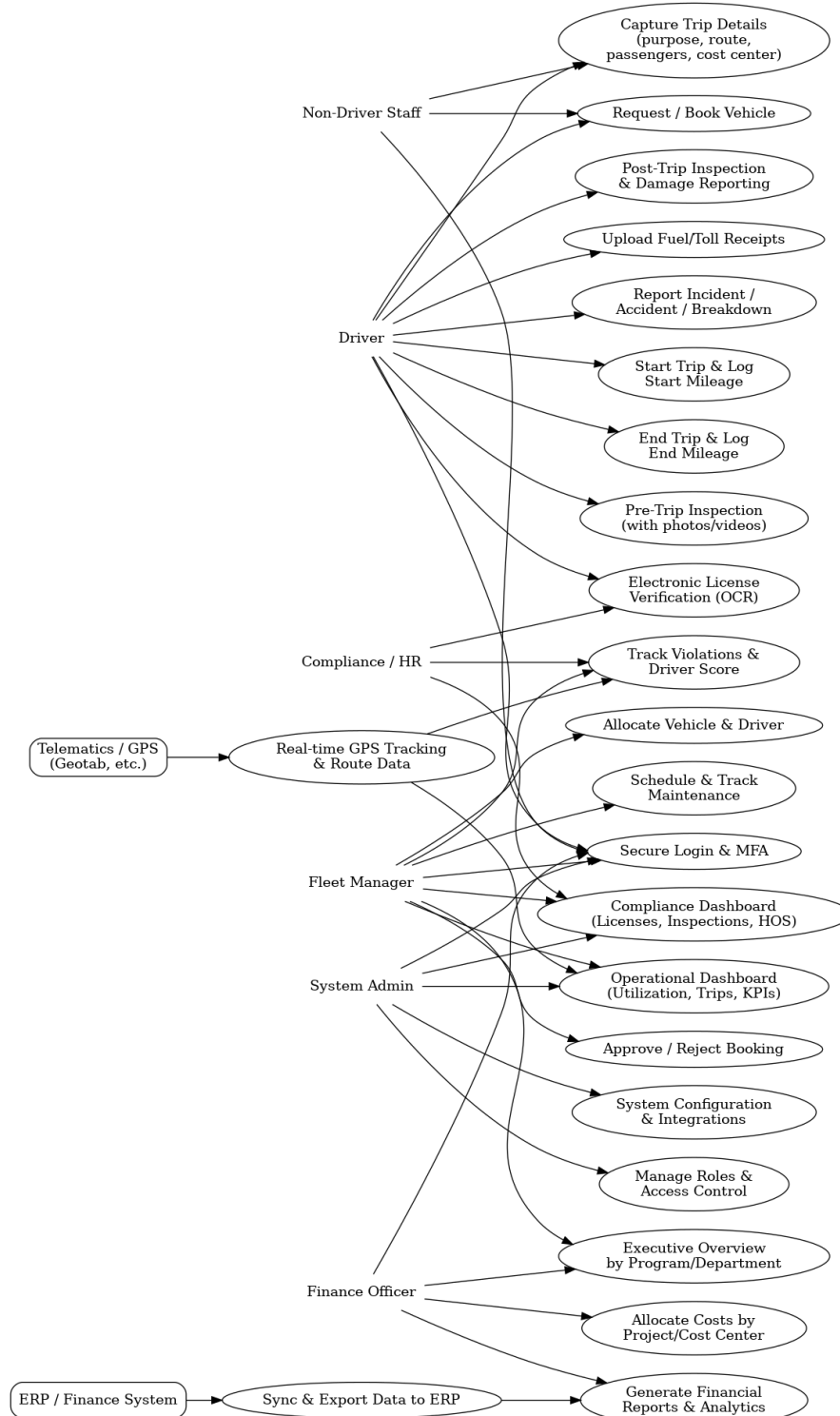


Figure 2. The diagram shows how each WVZ user role interacts with the system across bookings, trips, inspections, and reporting. It captures essential role-based workflows required for fleet transparency and accountability.

Unique Value Proposition

A unified, low-bandwidth fleet management system that digitises bookings, inspections, and trip logs while enabling real-time accountability and cost-centre visibility. Purpose-built for NGO workflows, it supports donor transparency, rural deployment, and seamless collaboration across WVZ teams.

Conclusion and Future Work

Impact Report

This project supports **SDG 9 (Industry, Innovation and Infrastructure)** by modernising WVZ's fleet infrastructure with automation and real-time data, and **SDG 11 (Sustainable Cities and Communities)** by improving transport safety, accountability, and service delivery. Overall, it strengthens WVZ's ability to reach vulnerable communities more reliably and sustainably.

Company Alignment

This project strengthens WVZ's mission by improving the fleet operations that support safe, timely outreach to vulnerable children and communities. By enhancing accountability and reducing operational delays, it reinforces WVZ's commitment to transparency and effective service delivery. Overall, it supports the vision of ensuring every child experiences life in all its fullness through reliable, data-driven operations.

Timeline and Work Plan

TABLE II. Implementation Plan Showing the Two-Month Pilot Timeline and Incremental Feature Development for Mota50.

Phase	Duration	Activities
Pilot Implementation	Dec - Jan	Deploy core features (bookings, trip logs, inspections, cost-centre reporting) at a selected WVZ site; gather feedback; refine workflows.
Next Prototype Features	Within Dec – Jan (Iterative Cycles)	Mobile app, geo-fencing, route monitoring, violation-point system, GPS validation, encryption in transit, GPS tracking, SMS/email gateways, IoT & telematics, waiver requests, notifications, calendar view, smart reminders, fleet overview.
Future Enhancements	Within 2-Months	Integrate AI/ML for automated insights, safety analytics, optimisation, and predictive fleet management.

References

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